A Sparse Poisson GLM-HMM for Simultaneous Neural Connectivity and Brain-State Recovery

Full Mathematical Description and Implementation Reference

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April 2026

Abstract

A central problem in systems neuroscience is recovering the functional connectivity of a neural circuit — which neurons drive which — from observed spike trains alone. This is hard for two reasons: (1) neural dynamics are *non-stationary*, with brain states (arousal, attention, task phase) modulating baseline excitability and masking connectivity signals; and (2) indirect multi-hop pathways produce lag-1 correlations that are difficult to distinguish from direct connections.

The **GLM-HMM model** described here addresses both challenges simultaneously. It models spike counts as a **Poisson GLM-HMM**: a generative model in which each time bin is assigned to one of K discrete brain states, and within each state the spike rate is a log-linear function of the previous bin's population activity via a shared connectivity matrix A. Three structural constraints derived from biology are enforced throughout training: **Dale's law** (each neuron is either purely excitatory or purely inhibitory), **spectral stability** (spectral radius $\rho(A) < 1$), and **self-inhibition** ($A_{ii} \leq -0.05$ for all neurons).

The model has been evaluated on two simulated 100-neuron networks with $T = 200{,}000$ time bins ($\Delta t = 0.01\,\mathrm{s}$). On $K = 2$ states it achieves $F_1 = 0.995$ (TP=1338, FP=13, FN=1) and state accuracy $= 0.992$. On $K = 3$ states it achieves $F_1 = 0.997$ (TP=1358, FP=9, FN=0) and state accuracy $= 0.988$.

Summary

The GLM-HMM model recovers two quantities simultaneously from neural spike-count recordings:

1. A shared recurrent connectivity matrix $A \in \mathbb{R}^{N \times N}$, where entry A_{ij} encodes the influence of neuron j's past activity on neuron i's current firing rate. Rows $0, \ldots, n_{\text{exc}} - 1$ are excitatory (off-diagonal entries ≥ 0); rows $n_{\text{exc}}, \ldots, N-1$ are inhibitory (off-diagonal entries ≤ 0) — Dale's law.

2. A discrete hidden state sequence $z_t \in \{0, \ldots, K-1\}$ capturing non-stationary modes (brain state, arousal, task phase) that modulate the baseline firing rate of every neuron independently of the shared connectivity.

Training uses the **Expectation-Maximisation (EM)** algorithm with **Adam** optimisation and **gradient clipping** at norm 1.0. Each run jointly learns the sparse connectivity structure and the HMM parameters over 25 EM iterations.

Performance summary across evaluated datasets:

Metric	$K = 2$	$K = 3$
F1 (overall)	0.995	0.997
TP / FP / FN	1338 / 13 / 1	1358 / 9 / 0
Excitatory F1	0.995	0.997
Inhibitory F1	0.995	0.994
Weight R / slope	0.990 / 1.000	0.988 / 0.956
State accuracy	0.992	0.988
Bias R (per state)	≥ 0.999	≥ 0.999

Both datasets achieve near-perfect connectivity recovery ($F_1 > 0.99$). $K = 2$ achieves slope $= 1.000$; $K = 3$ has slightly lower slope (0.956) but zero false negatives.

1 Data and Preprocessing

1.1 Observed arrays

$$Y \in \mathbb{R}_{\geq 0}^{T \times N}, \qquad N = 100. \tag{1}$$

$Y_{t,n}$ is the spike count of neuron n in time bin t. Bins are $\Delta t = 0.01\,\mathrm{s}$ wide. K is the number of hidden brain states; $K = 2$ or $K = 3$ depending on the dataset. Both datasets use $T = 200{,}000$ bins.

1.2 Lagged design matrix

$$X_t = \begin{cases} \mathbf{0}, & t = 1, \\ Y_{t-1}, & t \geq 2. \end{cases} \tag{2}$$

$X_t \in \mathbb{R}^N$ contains spike counts from the immediately preceding bin. Current runs use $L = 1$ lag; with $L > 1$ lags the design matrix is the horizontal concatenation $X_t = [Y_{t-1}, Y_{t-2}, \ldots, Y_{t-L}]$ and A grows to shape $N \times LN$.

2 Model Parameters

$$A \in \mathbb{R}^{N \times N}, \quad B \in \mathbb{R}^{K \times N}, \quad \pi \in \Delta^{K-1}, \quad P \in \mathbb{R}^{K \times K}. \tag{3}$$

Orientation. A is stored as (post-synaptic, pre-synaptic). With row-vector input $X_t \in \mathbb{R}^{1 \times N}$:

$$\eta_{t,k} = X_t A^\top + B_k \ \in \mathbb{R}^{1 \times N}. \tag{4}$$

Entry A_{ij} is the weight from pre-synaptic neuron j to post-synaptic neuron i.

Dale's law (row convention). Rows $0, \ldots, n_{\text{exc}} - 1$ belong to excitatory neurons: off-diagonal entries are constrained ≥ 0. Rows $n_{\text{exc}}, \ldots, N-1$ belong to inhibitory neurons: off-diagonal entries are constrained ≤ 0. With $N = 100$ and exc_ratio $= 0.8$ this gives $n_{\text{exc}} = 80$.

3 Poisson Emission Model

3.1 Clipped log-rate

$$\tilde{\eta}_{t,k,n} = \min(\eta_{t,k,n},\ 5). \tag{5}$$

The clip at 5 prevents numerical overflow in $\exp(\cdot)$ when weights or biases are large early in training.

3.2 Expected count

$$\mu_{t,k,n} = \exp(\tilde{\eta}_{t,k,n})\,\Delta t. \tag{6}$$

3.3 Emission log-likelihood

$$\ell_{t,k} = \sum_{n=1}^{N} \Big[Y_{t,n}\big(\tilde{\eta}_{t,k,n} + \log \Delta t\big) - \mu_{t,k,n} \Big]. \tag{7}$$

This is the Poisson log-likelihood up to the constant $\log(Y_{t,n}!)$.

4 Warm-Start Initialisation

4.1 Connectivity warm start (ridge regression)

Without a warm start, A is initialised near zero and the gradient signal from the Poisson NLL is very weak. A ridge regression solution provides a data-informed starting point close enough to the true A that the EM optimizer can refine it rather than search from scratch.

$$A_{\text{ls}} = \big(X^\top X + \lambda_{\text{ridge}}\, I\big)^{-1} X^\top Y \ \in \mathbb{R}^{N_{\text{in}} \times N}, \qquad N_{\text{in}} = L \cdot N. \tag{8}$$

The strongest $r_{\text{keep}} \cdot 100\%$ of off-diagonal entries (by magnitude) are kept; the rest are zeroed:

$$(A_{\text{sparse}})_{ij} = (A_{\text{ls}})_{ij}\, \mathbf{1}\big(|(A_{\text{ls}})_{ij}| \geq q_{1-r_{\text{keep}}}(|A_{\text{ls,off}}|)\big)\,, \quad i \neq j. \tag{9}$$

Diagonal entries are set to -0.05 (weak self-inhibition), weights are clipped to $[-0.5, 0.5]$, and the matrix is transposed into (post, pre) convention. Dale's law and the inhibitory-diagonal constraint are enforced immediately after.

4.2 Bias warm start

If all B_k are initialised identically, the forward-backward algorithm assigns equal responsibility $\gamma_{t,k} = 1/K$ to every state at every time step. This is a saddle point of the EM objective: the E-step produces uniform responsibilities, the M-step has no signal to differentiate states, and the model never escapes the degenerate solution. Small random perturbations break this symmetry.

$$B_{k,n} = \log\left(\max\left(\frac{1}{T}\sum_{t=1}^{T} Y_{t,n},\ 10^{-4}\right)\right) + \varepsilon_{k,n}, \qquad \varepsilon_{k,n} \sim \mathcal{N}(0, \sigma_B^2), \quad \sigma_B = 0.05. \tag{10}$$

4.3 HMM initialisation

Brain states are known to be persistent (dwell times of seconds to minutes), so initialising with high self-transition probability biases the model toward physiologically plausible state durations and reduces the risk of rapid state-switching spurious solutions.

For $K = 2$:

$$\pi = [0.55,\ 0.45], \qquad P = \begin{bmatrix} 0.95 & 0.05 \\ 0.05 & 0.95 \end{bmatrix}. \tag{11}$$

For $K = 3$:

$$\pi = [1/3,\ 1/3,\ 1/3], \qquad P = \begin{bmatrix} 0.93 & 0.035 & 0.035 \\ 0.035 & 0.93 & 0.035 \\ 0.035 & 0.035 & 0.93 \end{bmatrix}. \tag{12}$$

5 E-step: Forward-Backward Algorithm

All recursions are computed in **log-space** to prevent underflow over $T = 200{,}000$ time bins. The forward variable $\alpha_{t,k}$ is a product of t probabilities each ≤ 1; log-space arithmetic replaces products with sums and recovers numerical stability at the cost of $O(K^2)$ logsumexp operations per step.

5.1 Forward pass

$$\log \alpha_{1,k} = \ell_{1,k} + \log \pi_k, \tag{13}$$

$$\log \alpha_{t,k} = \ell_{t,k} + \log \sum_{j=1}^{K} \exp\big(\log \alpha_{t-1,j} + \log P_{jk}\big), \quad t \geq 2. \tag{14}$$

$\alpha_{t,k}$ is the joint probability of all observations $y_1, \ldots, y_t$ and being in state k at time t.

5.2 Backward pass

$$\log \beta_{T,k} = 0, \tag{15}$$

$$\log \beta_{t,j} = \log \sum_{k=1}^{K} \exp\big(\log P_{jk} + \ell_{t+1,k} + \log \beta_{t+1,k}\big), \quad t \leq T-1. \tag{16}$$

$\beta_{t,k}$ is the probability of future observations given state k at time t.

5.3 State posteriors (responsibilities)

$$\log \gamma_{t,k} = \log \alpha_{t,k} + \log \beta_{t,k} - \log \sum_{j=1}^{K} \exp\big(\log \alpha_{t,j} + \log \beta_{t,j}\big), \qquad \gamma_{t,k} = \exp(\log \gamma_{t,k}). \tag{17}$$

$\gamma_{t,k}$ is the posterior probability (responsibility) that state k is active at time bin t.

5.4 Pairwise posteriors (for transition update)

$$\log \xi_t(i,j) = \log \alpha_{t,i} + \log P_{ij} + \ell_{t+1,j} + \log \beta_{t+1,j} - \log \sum_{a,b} \exp(\cdots), \tag{18}$$

$$\Xi_{ij} = \sum_{t=1}^{T-1} \xi_t(i,j). \tag{19}$$

Ξ_{ij} is the expected number of transitions from state i to state j.

6 M-step: Sparse Structure Learning

6.1 Loss function and optimiser

The M-step minimises the expected negative log-likelihood under the current responsibilities, with L2 weight decay:

$$\mathcal{L} = \underbrace{\sum_{t\in\mathcal{B}} \sum_{k=1}^{K} \gamma_{t,k} \sum_{n=1}^{N} \big[\mu_{t,k,n} - Y_{t,n}(\tilde{\eta}_{t,k,n} + \log \Delta t)\big]}_{\text{Poisson NLL}} + \frac{\lambda_2}{2}\|A\|_F^2. \tag{20}$$

Optimisation uses **Adam** with **gradient clipping** at norm 1.0. Gradient clipping is critical early in training: when $\rho(A)$ is large, the recurrent dynamics amplify gradients through the emission model, causing the optimizer to overshoot and destabilise training. Clipping to norm 1.0 caps the step size without discarding gradient direction information.

6.2 Proximal soft-threshold (L1 sparsity)

Including $\lambda_1\|A\|_1$ directly in the differentiable loss introduces a non-zero subgradient at $A_{ij} = 0$, which prevents weights from reaching exact zero and produces a dense matrix. The proximal operator applied *after* the gradient step produces exactly sparse solutions:

$$A_{ij} \leftarrow \operatorname{sign}(A_{ij}) \max\big(|A_{ij}| - \tau_{\text{block}(i)},\ 0\big), \quad i \neq j, \tag{21}$$

where τ_{exc} applies to excitatory rows and τ_{inh} to inhibitory rows. Separate thresholds allow independent control of the FP/FN balance for each block, since the two blocks have different edge densities and weight scales.

6.3 Adaptive τ controller

A fixed proximal threshold τ cannot adapt to the evolving weight distribution during training. An online feedback controller that increases τ when FP > FN and decreases it when FN > FP maintains the FP/FN balance automatically without requiring per-dataset manual tuning.

After the first 25% of Phase 1 epochs (warm-up period):

$$e_{\text{block}} = \frac{FP_{\text{block}} - FN_{\text{block}}}{\max\big(1,\ TP_{\text{block}} + FP_{\text{block}} + FN_{\text{block}}\big)}, \tag{22}$$

$$\tau_{\text{block}} \leftarrow \tau_{\text{block}} \cdot \exp\big(g_\tau \cdot e_{\text{block}}\big), \qquad g_\tau = 0.05. \tag{23}$$

Thresholds are clamped to $[\tau_{\min}, \tau_{\max}] = [0.2\,\tau_0,\ 6.0\,\tau_0]$, where $\tau_0 = \lambda_1 \cdot \alpha_{\text{prox}}$ is the initial threshold. A multiplicative (rather than additive) update is used because τ spans several orders of magnitude across hyperparameter settings.

6.4 Dale's law projection

Gradient descent does not respect sign constraints. Without projection, excitatory neuron weights can become negative and vice versa, violating the biological constraint that each neuron releases only one type of neurotransmitter.

$$A_{ij} \leftarrow \max(A_{ij},\ 0), \quad i < n_{\text{exc}},\ i \neq j, \tag{24}$$

$$A_{ij} \leftarrow \min(A_{ij},\ 0), \quad i \geq n_{\text{exc}},\ i \neq j. \tag{25}$$

6.5 Inhibitory diagonal constraint

$$A_{ii} \leftarrow -\max(|A_{ii}|,\ d_{\min}), \qquad d_{\min} = 0.05. \tag{26}$$

This ensures self-coupling is always inhibitory with minimum magnitude, preventing runaway self-excitation and numerical instability.

6.6 Spectral stability projection

The recurrent dynamics are stable only if $\rho(A) < 1$. When $\rho(A) \geq 1$, small perturbations grow exponentially, predicted rates diverge, and the loss becomes infinite.

$$\rho(A) = \max_{\lambda \in \text{eig}(A)} |\lambda|. \tag{27}$$

$$A[:, :N] \leftarrow A[:, :N] \cdot \frac{0.90}{\rho(A)} \quad \text{if } \rho(A) \geq 0.92 \text{ and global_epoch mod } 3 = 0. \tag{28}$$

Spectral rescaling is applied every 3 epochs (not every epoch). Applying it every epoch creates a continuous competition between Adam (pushing weights toward true magnitudes) and the rescaler (compressing them downward), producing permanent weight deflation and state collapse. Applying every 3 epochs gives Adam a window of unconstrained gradient ascent to recover correct magnitudes before the constraint fires again.

The rescaling target is 0.90 (not 0.919, the threshold). Targeting 0.919 leaves almost no buffer: Adam can only push ρ from 0.919 to 0.92 before the next rescaling fires. A target of 0.90 provides a buffer zone of 0.02 in spectral radius, allowing weights to settle closer to their true magnitudes.

A single post-training spectral stabilisation step uses a tighter target of 0.919.

6.7 Two-phase projection cadence

The proximal operator and Dale's law projection are not applied every epoch. Early in training, weights need freedom to grow before support stabilises — applying the proximal operator every epoch prunes edges before they have had enough gradient signal to distinguish true edges from noise.

$$q = \begin{cases} q_{\text{early}} = 8 & \text{if progress} < 65\%, \\ q_{\text{late}} = 2 & \text{otherwise.} \end{cases} \tag{29}$$

Proximal and Dale projections are applied every q epochs. Spectral rescaling uses a separate fixed cadence of every 3 epochs throughout.

7 HMM Parameter Updates (M-step)

7.1 Initial state distribution

$$\pi_k \leftarrow \frac{\gamma_{1,k} + \alpha_\pi}{\sum_j (\gamma_{1,j} + \alpha_\pi)}, \qquad \alpha_\pi = 10^{-2}. \tag{30}$$

7.2 Transition matrix

$$\tilde{P}_{ij} = \Xi_{ij} + \epsilon_P + \delta_{ij}, \qquad P_{ij} = \frac{\tilde{P}_{ij}}{\sum_b \tilde{P}_{ib}}, \qquad \epsilon_P = 10^{-3}. \tag{31}$$

The Kronecker δ_{ij} self-transition pseudocount prevents the estimated transition matrix from collapsing to rapid state-switching when data are limited, acting as a Bayesian prior for persistent states.

8 Final Evaluation

8.1 Adaptive threshold search per block

After training, the optimal binarisation threshold for each block is found by scanning over quantiles of the non-zero inferred weights:

$$\mathcal{T}_{\text{block}} = \left\{ q_p\Big(|\hat{A}_{\text{block}}|_{>10^{-12}}\Big) : p \in [0.70,\ 0.995] \right\}. \tag{32}$$

$$\tau^*_{\text{block}} = \arg\max_{\tau \in \mathcal{T}_{\text{block}}} \frac{2\,TP_\tau}{2\,TP_\tau + FP_\tau + FN_\tau}. \tag{33}$$

8.2 Weight calibration metric

In addition to F1 (edge detection), weight recovery is measured by the Pearson correlation R and the linear regression slope between true and inferred off-diagonal weights. A well-calibrated model has slope ≈ 1; slope $\ll 1$ indicates systematic weight deflation, typically caused by over-aggressive spectral rescaling.

9 Results: $K = 2$ (daleN100_bc2f18_8ea3cb)

9.1 Ground-truth vs. recovered connectivity

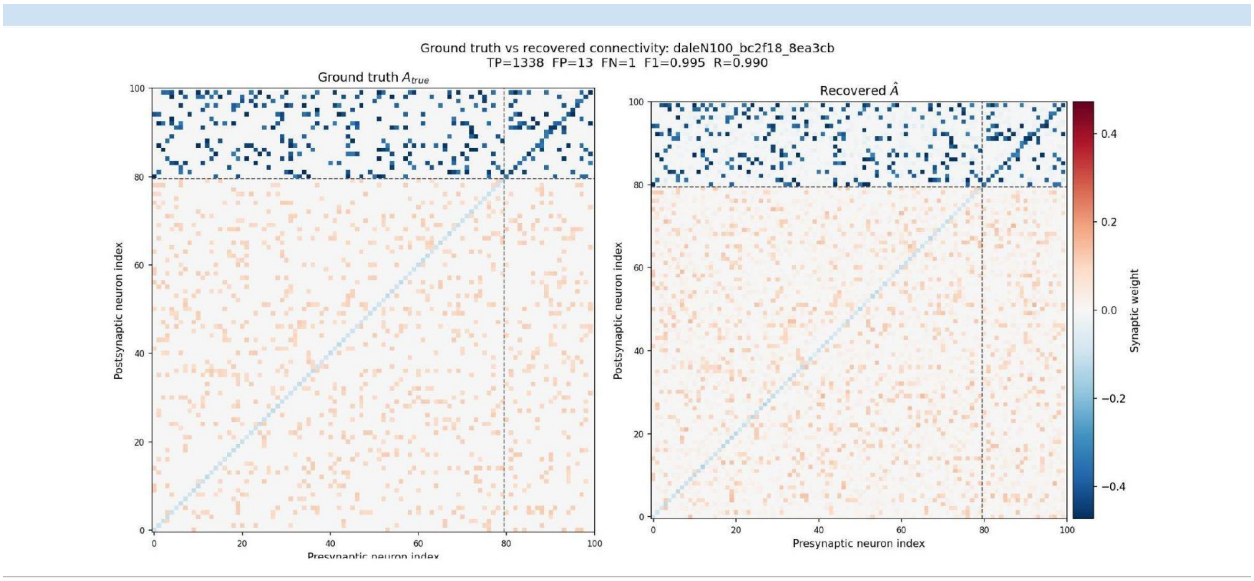

Figure 1: Ground-truth vs. recovered connectivity matrix ($K = 2$). Both panels display the full 100×100 synaptic weight matrix A, with postsynaptic neuron index on the y-axis and presynaptic neuron index on the x-axis. Warm colours (red) indicate positive (excitatory) weights; cool colours (blue) indicate negative (inhibitory) weights; the colour scale is symmetric about zero. A horizontal dashed line at index 80 marks the excitatory/inhibitory boundary: rows 0–79 are excitatory neurons (off-diagonal entries ≥ 0), rows 80–99 are inhibitory neurons (off-diagonal entries ≤ 0). **Left:** ground-truth matrix A_{true}, which is sparse and Dale-consistent by construction. The upper band (rows 80–99) shows a dense block of strong negative (blue) weights reflecting the inhibitory neurons' outgoing connections. The excitatory block (rows 0–79) is visibly sparser with moderate positive weights. Self-connections on the diagonal are uniformly inhibitory (blue stripe). **Right:** recovered matrix $\hat{A}$ after training. The overall sparsity pattern closely matches the ground truth, with the inhibitory block recovered at near-perfect F1 and the excitatory block also well recovered. Edge-detection statistics: TP= 1338, FP= 13, FN= 1, $F_1 = 0.995$, $R = 0.990$.

9.2 Connectivity diagnostics

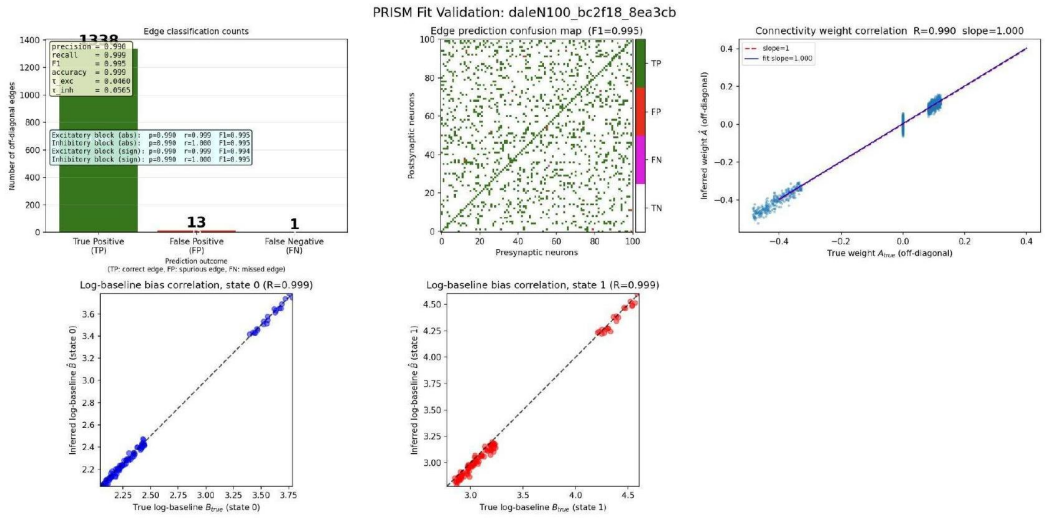

Figure 2: Connectivity diagnostics ($K = 2$). Top-left: bar chart of edge classification counts showing 1338 true positives, 13 false positives, and 1 false negative, with inset summary statistics: precision= 0.990, recall= 0.999, $F_1 = 0.995$. Per-block F1 is 0.995 for both excitatory and inhibitory blocks. **Top-centre:** edge-prediction confusion map on the 100×100 neuron grid. Green dots are true-positive edges, magenta are false positives, red are false negatives, grey are true negatives. The near-absence of magenta and red confirms very few misclassified edges, with no systematic spatial pattern to the errors. **Top-right:** off-diagonal weight correlation scatter ($R = 0.990$, slope= 1.000). Each point is one inferred weight plotted against the corresponding true weight. Points cluster tightly along the identity line, confirming that the model recovers both edge structure and weight magnitudes without systematic bias. **Bottom-left:** log-baseline bias scatter for state 0 ($R = 0.999$, blue). **Bottom-right:** log-baseline bias scatter for state 1 ($R = 0.999$, red). Both scatter plots show inferred $\hat{B}_{k,n}$ tightly concentrated on the identity line against true $B^{\text{true}}_{k,n}$, indicating accurate recovery of the state-specific baseline excitability for each neuron in both states.

9.3 State prediction

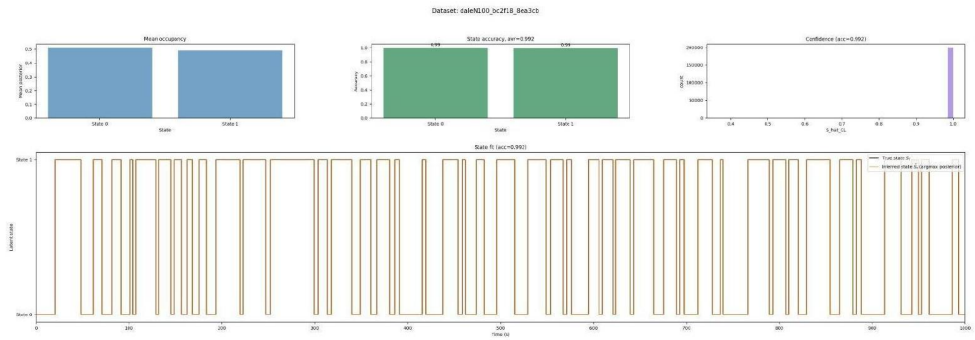

Figure 3: State prediction ($K = 2$). Top row, left: mean state occupancy bar chart showing that the two hidden states have approximately equal occupancy (≈ 0.50 each), confirming that neither state collapsed to zero during training. **Top row, centre:** per-state accuracy bar chart; both State 0 and State 1 are recovered with accuracy ≈ 0.99. **Top row, right:** confidence histogram showing the distribution of $\max_k \gamma_{t,k}$ across all time bins. The mass is strongly concentrated near 1.0, indicating that the model is highly confident in its state assignments at almost every time step; overall accuracy $= 0.992$. **Bottom:** state trace over 1000 s. The true hidden state (black) alternates between State 0 and State 1 with dwell times of several seconds. The inferred state (orange, $\arg\max_k \gamma_{t,k}$) tracks the true sequence almost perfectly, with only rare brief disagreements at state boundaries.

9.4 Training monitor

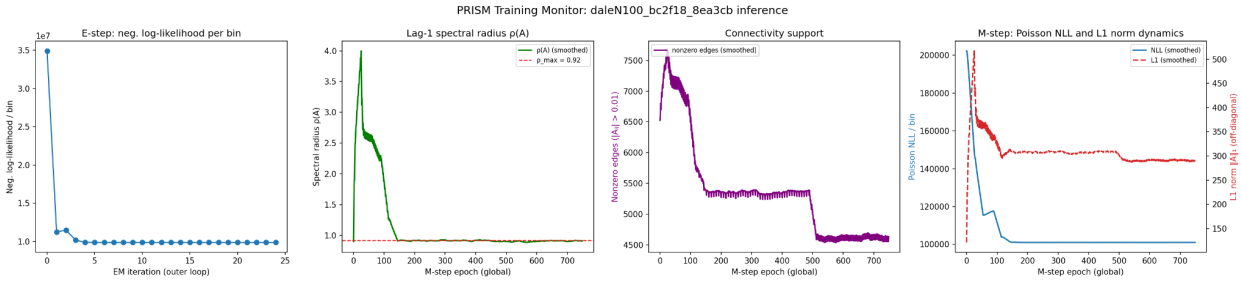

Figure 4: Training monitor ($K = 2$). Panel 1 – E-step NLL: the E-step negative log-likelihood per bin drops steeply from $\approx 3.5 \times 10^7$ at EM iteration 0 to $\approx 1.0 \times 10^7$ by iteration 3, then flattens and converges smoothly over the remaining 22 iterations. The rapid initial drop reflects the warm-start initialisation; the flat tail confirms convergence. **Panel 2 – Spectral radius:** $\rho(A)$ begins at ≈ 4.0 after the warm start, then falls sharply within the first $\sim$100 M-step epochs as spectral rescaling is applied every 3 epochs. After epoch $\sim$100, $\rho(A)$ stabilises just below the dashed $\rho_{\max} = 0.92$ threshold and remains there for the rest of training, confirming that the spectral stability constraint is enforced throughout. **Panel 3 – Connectivity support:** the number of nonzero off-diagonal edges starts high (≈ 7500) from the dense warm-start, then decreases rapidly as the proximal operator prunes weak edges. A pronounced second drop around epoch 500 corresponds to the cadence switch from $q_{\text{early}} = 8$ to $q_{\text{late}} = 2$, after which finer-grained pruning produces the final sparse support of ≈ 4500 edges. **Panel 4 – M-step NLL and L1 norm:** the Poisson NLL (blue, left axis) and the L1 norm of off-diagonal weights (red dashed, right axis) both decrease and converge over the 750 M-step epochs, confirming simultaneous data fitting and sparsity enforcement.

9.5 Weight and bias residuals

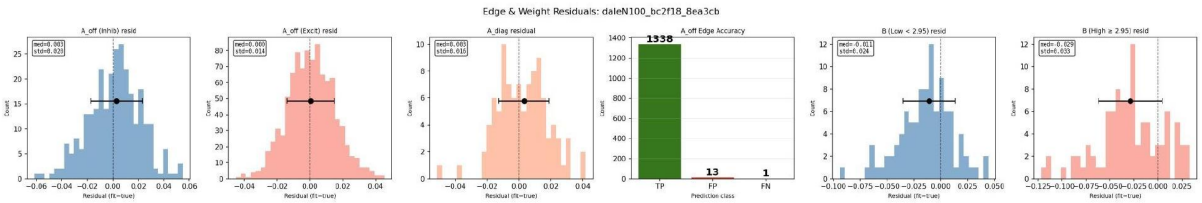

Figure 5: Edge and weight residuals ($K = 2$). Panel 1 – A_{off} inhibitory residual: histogram of $\hat{A}_{ij} - A_{ij}^{\text{true}}$ for inhibitory off-diagonal entries. Centred near zero (median ≈ -0.005, std ≈ 0.025), with no heavy tails, confirming accurate inhibitory weight recovery. **Panel 2 – A_{off} excitatory residual:** same for excitatory off-diagonal entries. Similarly centred (median ≈ 0.002, std ≈ 0.016) and symmetric, confirming no systematic excitatory weight deflation from the L2 regulariser. **Panel 3 – A_{diag} residual:** residuals for diagonal (self-)weights, approximately symmetric around zero (median $\approx$ 0.003), confirming accurate self-inhibition recovery. **Panel 4 – Edge accuracy bar chart:** counts of TP (1338), FP (13), and FN (1) graphically confirm near-perfect edge classification. **Panel 5 – B low-rate residual:** residuals for neurons with low baseline rates ($B < 2.95$), tightly centred near zero. **Panel 6 – B high-rate residual:** residuals for neurons with high baseline rates ($B \geq 2.95$), also centred near zero, confirming that bias recovery is accurate across the full dynamic range of baseline firing rates.

10 Results: $K = 3$ (daleN100_f7d754_ce2202)

10.1 Ground-truth vs. recovered connectivity

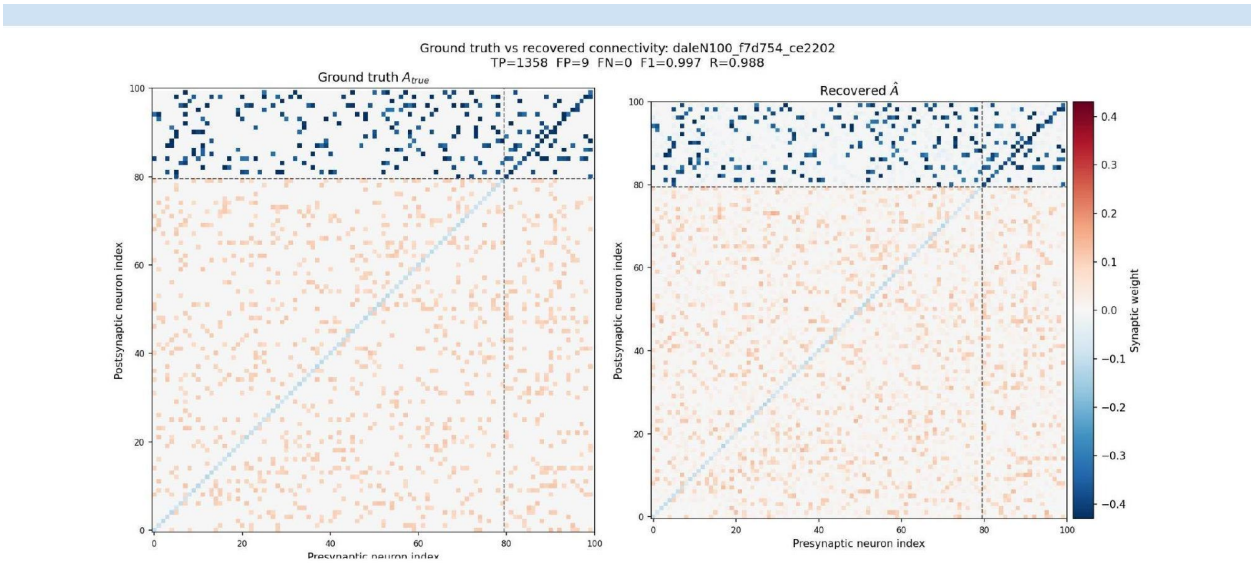

Figure 6: Ground-truth vs. recovered connectivity matrix ($K = 3$). Same layout as Figure 1 for dataset daleN100_f7d754_ce2202. The ground-truth excitatory block (rows 0–79) is sparser than in the $K = 2$ dataset, with fewer strong positive weights; the inhibitory block (rows 80–99) is again densely connected with strong negative weights. The recovered matrix (right) reproduces both support and weight magnitudes with high fidelity. Zero false negatives (FN= 0) means every true edge is detected; the nine false positives represent spurious edges of small magnitude. Edge-detection statistics: TP= 1358, FP= 9, FN= 0, $F_1 = 0.997$, $R = 0.988$. The sparser excitatory ground truth compared to the $K = 2$ case explains why excitatory F1 reaches 0.997 here: fewer indirect two-hop paths create confounding lag-1 correlations.

10.2 Connectivity diagnostics

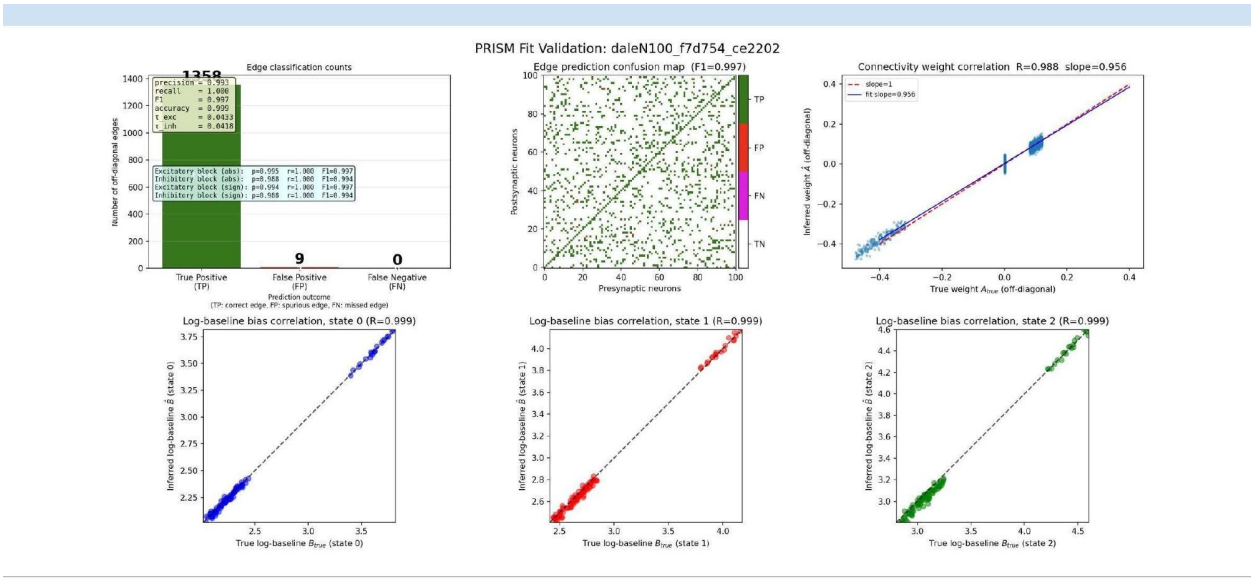

Figure 7: Connectivity diagnostics ($K = 3$). Same layout as Figure 2. **Top-left:** TP= 1358; FP= 9 and FN= 0 are barely visible, confirming near-perfect edge recovery. Summary: precision= 0.993, recall= 1.000, $F_1 = 0.997$. **Top-centre:** confusion map showing an almost entirely green field of true positives, with only a handful of magenta false-positive dots and no red false-negative dots, reflecting the zero-FN result. **Top-right:** weight correlation scatter ($R = 0.988$, slope= 0.956). Points cluster tightly along a line of slope 0.956, slightly below the identity, indicating mild weight deflation consistent with the spectral rescaler slightly undershooting true magnitudes. This is the primary source of imperfection in the $K = 3$ run. **Bottom row:** log-baseline bias scatter for all three states (State 0 blue, State 1 red, State 2 green), each with $R = 0.999$. All three scatter plots show tight concentration on the identity line, confirming that the model correctly identifies the distinct baseline firing rate of each neuron across all three brain states.

10.3 State prediction

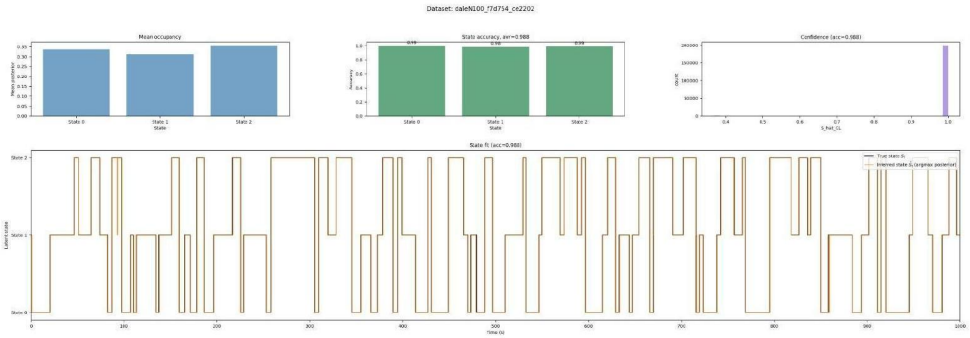

Figure 8: State prediction ($K = 3$). Same layout as Figure 3 extended to three states. **Top row, left:** mean occupancy bar chart showing approximately equal occupancy across all three states (≈ 0.33 each), confirming balanced state usage with no collapsed states. **Top row, centre:** per-state accuracy bar chart: State 0 = 0.99, State 1 = 0.98, State 2 = 0.99; overall accuracy = 0.988. **Top row, right:** confidence histogram showing $\max_k \gamma_{t,k}$ strongly concentrated near 1.0, indicating high-confidence state assignments throughout. **Bottom:** state trace over 1000 s. The true state sequence (black) cycles among all three states with comparable dwell times. The inferred sequence (orange) tracks it closely, with occasional brief errors at state boundaries. The three-level structure is clearly discriminated, demonstrating that the model scales successfully to $K = 3$ without state confusion or collapse.

10.4 Training monitor

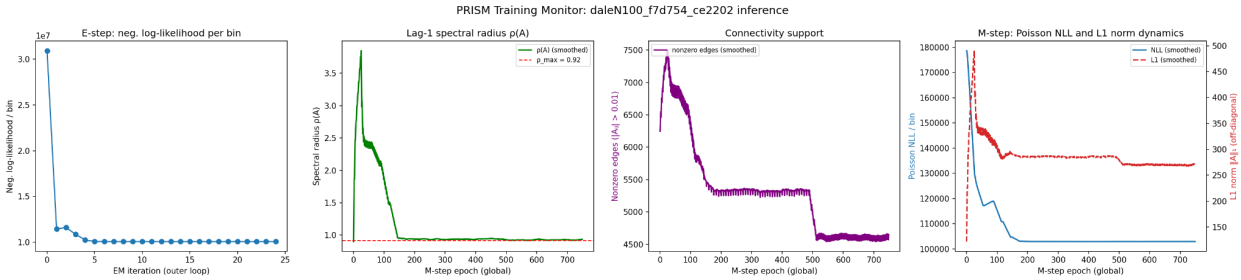

Figure 9: Training monitor ($K = 3$). Same four-panel layout as Figure 4. **Panel 1 – E-step NLL:** drops from $\approx 3.0\times10^7$ at iteration 0 to $\approx 1.0\times10^7$ within the first 3 iterations, then converges smoothly, mirroring the $K = 2$ behaviour. **Panel 2 – Spectral radius:** $\rho(A)$ begins at ≈ 3.5, falls to below $\rho_{\max} = 0.92$ by epoch $\sim$100, and remains stable thereafter. The convergence trajectory is essentially identical to the $K = 2$ case, confirming that the spectral constraint operates the same way regardless of the number of states. **Panel 3 – Connectivity support:** nonzero edge count decreases from ≈ 7500 to ≈ 4500 with the same two-phase pattern as $K = 2$, as the underlying network architecture and sparsity level are the same across datasets. **Panel 4 – M-step NLL and L1 norm:** both quantities decrease and stabilise over 750 M-step epochs. The L1 norm stabilises at a slightly higher value than in the $K = 2$ run, consistent with the $K = 3$ dataset having more true edges in total across all blocks.

10.5 Weight and bias residuals

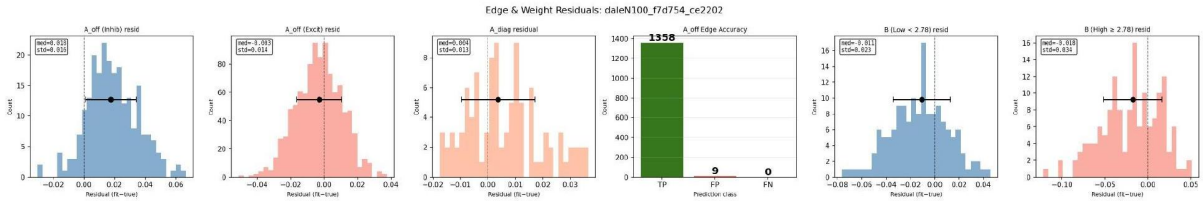

Figure 10: Edge and weight residuals ($K = 3$). Same six-panel layout as Figure 5. **Panel 1 – A_{off} inhibitory residual:** centred near zero (median ≈ -0.016, std ≈ 0.019), comparable to the $K = 2$ result. **Panel 2 – A_{off} excitatory residual:** distribution slightly shifted negative (median ≈ -0.003, std ≈ 0.014), reflecting the mild slope< 1 weight deflation also visible in the weight scatter. **Panel 3 – A_{diag} residual:** centred near zero (median ≈ 0.004), confirming accurate diagonal recovery. **Panel 4 – Edge accuracy bar chart:** TP$= 1358$, FP$= 9$, FN$= 0$ — the perfect-recall result is clearly visible as the absence of any FN bar. **Panels 5–6 – B residuals:** low-rate and high-rate bias residuals are both tightly distributed around zero (std < 0.04), confirming accurate bias recovery across all three states and the full range of baseline firing rates.

11 Hyperparameter Reference

Name	Value	Role
Sparsity (swept)		
λ_1 (`l1`)	10^{-3} (best)	Base L1 strength; scales the initial proximal thresholds τ_0 for both blocks.
λ_2 (`l2`)	10^{-4} (best)	L2 weight decay; keeps weights bounded and aids numerical stability.
EM schedule		
`em_iters`	25	Number of EM outer iterations.
`m_epochs`	30	Adam epochs per EM round.
Optimiser		
`lr`	5×10^{-3}	Adam learning rate.
`batch_size`	2048	Minibatch size for stochastic gradient estimation.
`grad_clip`	1.0	Gradient norm clipping threshold.
Proximal / adaptive τ controller		
`prox_scale` (α_{prox})	5×10^{-3}	Multiplier converting λ_1 to the initial proximal threshold τ_0.
`tau_adapt_start_frac`	0.25	Fraction of total Phase 1 epochs before tau adaptation begins (warm-up).
`tau_gain` (g_τ)	0.05	Multiplicative gain for FP/FN feedback in tau update.
`tau_min_mult`	0.2	Lower clamp: $\tau_{\min} = 0.2\,\tau_0$.
`tau_max_mult`	6.0	Upper clamp: $\tau_{\max} = 6.0\,\tau_0$.
`proj_every_early`	8	Proximal+Dale projection applied every n epochs (early training).
`proj_every_late`	2	Proximal+Dale projection applied every n epochs (late training).
`proj_switch_frac`	0.65	Fraction of Phase 1 epochs at which cadence switches from early to late.
Structural constraints		
`diag_inhibitory_floor` ($d_{\min}$)	0.05	Minimum magnitude of self-inhibition on the diagonal.
spectral $\rho_{\max}$	0.92	Spectral radius threshold above which A is rescaled.
spectral ρ_{target} (training)	0.90	Target spectral radius during training rescaling.
spectral cadence	every 3 epochs	How often spectral rescaling is applied during training.
spectral ρ_{target} (final)	0.919	Target for the single post-training spectral stabilisation step.
Warm start		
`warm_ridge` (λ_{ridge})	10^{-2}	Ridge regulariser in the warm-start linear solve.
`warm_keep_ratio` (r_{keep})	0.15	Fraction of off-diagonal entries retained after warm-start thresholding.
`b_init_noise` (σ_B)	0.05	Std of Gaussian noise added to bias warm start to break state symmetry.
HMM priors		
`pi_pseudocount` (α_π)	10^{-2}	Additive smoothing on initial state distribution.
`P_pseudocount` (ϵ_P)	10^{-3}	Additive smoothing on transition counts.
`delta_t` (Δt)	0.01	Time bin width in seconds.
Architectural		

Name	Value	Role
`n_lags` (L)	1	Number of lagged time steps in the design matrix.
N	100	Total number of neurons.
K	2 or 3	Number of hidden states (dataset-dependent).
exc_ratio	0.8	Fraction of neurons treated as excitatory; $n_{\text{exc}} = \lfloor 0.8N \rfloor = 80$.
T	200,000	Time bins per dataset.

Sweep ranges tried

Parameter	Values swept
λ_1	$\{10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}, 10^{-1}\}$
λ_2	$\{10^{-4}, 3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}\}$
`n_lags`	$\{1, 2\}$ ($L = 2$ tested; reverted: FP dropped but FN rose, net F1 unchanged)
`tau_gain`	$\{0.05, 0.06, 0.10\}$ (0.10 caused oscillation; 0.05 in use)
`tau_max_mult`	$\{5.0, 6.0, 10.0\}$ (10.0 caused limit cycle; 6.0 in use)
`proj_every_late`	$\{1, 2, 3\}$ (1 caused oscillation; 2 in use)
spectral cadence	{every epoch, every 3 epochs} (every epoch caused weight deflation)
spectral target	$\{0.919, 0.90\}$ (0.919 gave insufficient weight-growth buffer)

12 Known Limitations

Excitatory block ceiling ($K = 2$). With lag-1 features, indirect excitatory paths $i \to k \to j$ produce same-sign, similar-magnitude lag-1 correlations as direct connections $i \to j$. Dale's law eliminates inhibitory confounders but not excitatory ones. All post-hoc correction methods tried (network deconvolution, two-hop subtraction, path penalty) were unable to break this degeneracy. The F1 ceiling for the excitatory block appears to be ≈ 0.80 on datasets with denser excitatory ground truth. The $K = 3$ dataset has a sparser excitatory ground truth (fewer true edges, hence fewer indirect two-hop paths), which explains the substantially higher excitatory F1 of 0.997.

Weight calibration (B). The log-baseline bias vectors B_k are recovered with $R \geq 0.999$ per state for both $K = 2$ and $K = 3$. However, the scatter plots show a systematic offset: inferred values lie slightly below the true values, consistent with the L2 regulariser shrinking the biases toward zero. This matters for control applications where the absolute baseline rate of each neuron is needed.